

XINLU LIU

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EDUCATION

Fudan University

Sep 2021 - Jan 2024 (Expected)

Laboratory of Audio and Music Technology

M.Eng of *Computer Science*; GPA: 3.41/4.0

Huazhong University of Science and Technology

Sep 2017 - Jun 2021

B.Eng in *Automation*; Percentage: 86.59/100

RESEARCH EXPERIENCE

Online beat and downbeat tracking

Sep 2022 - Dec 2022

Literature research, Model design, Coding, Article writing

Shanghai, China

- Based on the structural characteristics of music, an online beat tracking framework with a "tilted pyramid" structure is designed. The framework improves upon the existing causal dilated model by introducing a small amount of latency at each layer, which significantly improves the results of real-time beat and downbeat tracking. The F-measure results for beat and downbeat are approximately 8% and 15% higher than those of the previous SOTA model, respectively.

Music source separation

Sep 2022 - Dec 2022

Literature research, Coding, visualization, Musical interpretation, Article writing

Shanghai, China

- A multi-scale Stripe-Transformer model is proposed in this study for music source separation. This model is capable of extracting stripe features for harmonics and drum beats in the spectrogram with fewer parameters while achieving SOTA performance.

Beat and downbeat tracking

Jan 2023 - March 2023

Literature research, Dataset collection, Article writing

Shanghai, China

- A hybrid network *XBeat* is proposed for beat and downbeat recognition. The *XBeat* network combines the characteristics of CNN and Transformer through feature exchange units, and introduces a novel data augmentation method. Experimental results demonstrate that *XBeat* can outperform traditional TCN baselines on most available public datasets.

Vocal register recognition

Aug 2019 - Dec 2020

data collection and processing, model design, GUI design

Wuhan, China

- A dataset for vocal register recognition is recorded and processed. A one-dimensional convolutional neural network (1D-CNN) is designed, which takes the Mel-spectrogram as input and achieved an accuracy of 99.15% on the self-constructed dataset, surpassing the performance of traditional machine learning models such as SVM.

PUBLICATIONS

Xinlu Liu, Jiale Qian, Qiqi He, Yi Yu, Wei Li: "LC-Beating: An Online System for Beat and Downbeat Tracking using Latency-Controlled Mechanism." ICME 2023

Jiale Qian, Xinlu Liu, Yi Yu and Wei Li: "Stripe-Transformer: deep stripe feature learning for music source separation." EURASIP JASMP.

Yuxuan Wu, Yifan He, Xinlu Liu, Yi Wang, Roger Dannenberg: "TransPlayer: Timbre Style Transfer with Flexible Timbre Control." ICASSP 2023.

Two papers submitted to CSMT 2023

COURSEWORKS

Postgraduate coursework: Audio Information Processing, Neural Networks and Deep Learning, Fundamentals and Applications of Generative Models, Machine Learning

Undergraduate coursework: Digital Signal Processing, Data Structure and Algorithm

SKILLS

Computer language	Python, C++, C, Matlab
Instruments	Guitar, Bass, Popular singing
Musical skills	Cubase, Mixing, Harmony
Deep learning	Pytorch, Keras